

Scaling Book Exercises: Section 12, Quiz 3

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Quiz: Beyond the Node Level

Source scan: ../handwritten/section_12_handwritten.pdf, pages 5–6

Reference: <https://jax-ml.github.io/scaling-book/gpus/>

Question 1: Fat-tree topology

The reference H100 SuperPod has 1024 GPUs, or 128 eight-GPU nodes. Each node connects to the leaf layer using eight 400 Gbit/s links:

$$8 \cdot \frac{400}{8} = \boxed{400 \text{ GB/s per node}}.$$

A scalable unit (SU) contains 32 nodes and eight leaf switches. The node-facing half of the 64 ports on each leaf switch provides

$$8 \cdot 32 \cdot \frac{400}{8} = \boxed{12.8 \text{ TB/s per SU}},$$

again 400 GB/s for each of its 32 nodes.

The leaf-to-spine links provide the same SU bandwidth:

$$8 \text{ leaf} \cdot 16 \text{ spine} \cdot 2 \text{ links} \cdot \frac{400}{8} = \boxed{12.8 \text{ TB/s per SU}}.$$

Finally, the 16 spine switches each support

$$64 \cdot \frac{400}{8} = 3.2 \text{ TB/s},$$

so their aggregate capacity is

$$16(3.2) = \boxed{51.2 \text{ TB/s}}.$$

Bisecting the 128 nodes leaves 64 nodes on each side. Counting both directions,

$$2 \cdot 64 \cdot 400 \text{ GB/s} = \boxed{51.2 \text{ TB/s}}.$$

Every level supplies 400 GB/s per node, so no level oversubscribes the others: the network has full bisection bandwidth.

Question 2: Scaling the pod

2048 GPUs. Keep the 32-node SU unchanged, increase from four to eight SUs, and double the spine layer from 16 to 32 switches. A leaf switch has only 64 ports, so use one 400 Gbit/s link from each leaf to each spine rather than two; doubling the number of spine switches preserves total bandwidth.

4096 GPUs. The leaf/spine design then runs out of ports. Add another fat-tree level of core switches. One reference arrangement uses

128 spine switches and 64 core switches.
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The essential principle is to add switching capacity at the same rate as node bandwidth so that the larger fabric remains non-oversubscribed.